2016.0 RANGE ROVER (LG), 204-00

SUSPENSION SYSTEM – GENERAL INFORMATION

DIAGNOSIS AND TESTING

PRINCIPLE OF OPERATION

For a detailed description of the suspension system and operation, refer to the relevant description and operation section of the workshop manual.REFER to:

Front Suspension (204-01 Front Suspension, Description and Operation),
Rear Suspension (204-02 Rear Suspension, Description and Operation).

INSPECTION AND VERIFICATION

WARNING:

Before carrying out a road test, make sure the vehicle is safe to do so. Failure to follow this instruction may result in personal injury

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault and may also cause additional faults in the vehicle being checked and/or the donor vehicle

NOTE:

Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests

1. Gather as much information from the driver as possible and verify the customer concern by carrying out a road test, as closely as possible reproducing the conditions under which the fault occurs

2. Visually inspect for obvious signs of mechanical damage

Visual Inspection

MECHANICAL
▪ Tire pressures ▪ Damaged wheels or tires ▪ Wheel bearing(s) ▪ Loose or damaged front or rear suspension components ▪ Loose, damaged or missing suspension fastener(s) ▪ Damaged or leaking air suspension components ▪ Worn or damaged suspension bushing(s) ▪ Loose, worn or damaged steering system components ▪ Damaged axle components ▪ Damaged chassis

3. If an obvious cause for an observed or reported condition is found, correct the cause (if possible) before proceeding to the symptom chart

4. If the cause is not visually evident, verify the symptom and refer to the symptom chart, alternatively check for diagnostic trouble codes (DTCs) and refer to the DTC index

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Crabbing	▪ Incorrect rear thrust angle ▪ Front or rear suspension components	▪ Check the rear alignment ▪ Check the front and rear suspension for signs of damage or wear
Drift/pull/wander	▪ Tire pressures ▪ Uneven tire wear ▪ Damaged steering components ▪ Wheel alignment ▪ Brake drag ▪ Unevenly loaded or overloaded vehicle	▪ Check and adjust the tire pressures (see visual inspection) ▪ Check for uneven tire wear, investigate the cause and rectify as necessary ▪ Check the steering for wear/damage ▪ Check and adjust the wheel alignment as necessary ▪ Check for binding brakes, rectify as necessary ▪ Advise the driver of the load issues
Front bottoming or riding low	▪ Damaged suspension components ▪ Air spring fault	▪ Check the suspension components for damage ▪ Check the dynamic suspension
Uneven tire wear	▪ Incorrect tire pressure (rapid center rib or inner and outer edge wear) ▪ Incorrect front or rear toe (rapid inner or outer edge wear) ▪ Incorrect camber (rapid	▪ Check and adjust the tire pressures (see visual inspection) ▪ Check and adjust the wheel alignment as necessary ▪ Balance the wheels and tires as necessary

	inner or outer edge wear) ▪ Tires out of balance (tires cupped or dished)	
Harsh ride	▪ Damaged suspension components ▪ Air spring fault	▪ Check the suspension components for damage ▪ Check the dynamic suspension
Shimmy or wheel tramp	▪ Wheels/tires ▪ Loose wheel nut(s) ▪ Loose front suspension fasteners ▪ Front wheel bearing(s) fault ▪ Worn or damaged suspension component bushing ▪ Loose, worn or damaged ball joint(s) ▪ Loose, worn or damaged steering components ▪ Front wheel alignment	▪ Check the wheels and tires for condition and balance ▪ Check and tighten the wheel nuts and suspension fasteners to specification ▪ Check the front wheel bearings, suspension bushings, ball joints and steering components for wear or damage ▪ Check and adjust the wheel alignment as necessary
Poor return ability of the steering (self-centering)	▪ Steering column ▪ Ball joints ▪ Steering components	▪ Check the steering column universal joints, etc ▪ Check the ball joints and other steering components
Sway or roll	▪ Loose front or rear stabilizer bar ▪ Worn lower suspension arm stabilizer bar insulators ▪ Air spring fault	▪ Check the stabilizer bar security and condition ▪ Rectify as necessary ▪ Check the function of the active stabilization system (where installed) ▪ Check the air springs
Vehicle leans to one side	▪ Front or rear suspension components	▪ Check the front and rear suspension ▪ Check the air springs

	▪ Air spring fault	
Evidence of fluid on rear shock absorber	▪ Fluid on shock absorber from an external source ▪ Fluid leaking from shock absorber	▪ Shock absorber not faulty, do not renew ▪ GO to Pinpoint Test A.

PINPOINT TESTS

NOTES:

- Residual oil left over from the shock absorber assembly process may create oil staining on the shock absorber tube. This will not affect the function of the shock absorber.
- Slight seepage is considered normal.

PINPOINT TEST A : SHOCK ABSORBER FLUID LEAK DIAGNOSIS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
A1: ASSESS LEAK	
	1 Assess the extent of the oil leakage
	Is the leakage serious enough to indicate that the shock absorber seal has failed? **Yes** GO to Pinpoint Test B. **No** Shock absorber not faulty, do not renew

PINPOINT TEST B : CONFIRM LEAK	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
B1: ROAD TEST	
	1 Clean all traces of oil from the shock absorber

	2 Drive the vehicle over a speed bump or similar ten times
	Is any fluid visible on the outside of the shock absorber? **Yes** Install a new shock absorber **No** Shock absorber not faulty, do not renew

DTC INDEX

For a list of diagnostic trouble codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00. REFER to: (100-00 General Information)

Diagnostic Trouble Code Index: Integrated Suspension Control Module (Description and Operation),
Diagnostic Trouble Code Index - DTC: Body Control Module (BCM) (Description and Operation),
Diagnostic Trouble Code Index - DTC: Power Steering Control Module (Description and Operation),
Diagnostic Trouble Code Index - DTC: Steering Angle Sensor Module (Description and Operation).